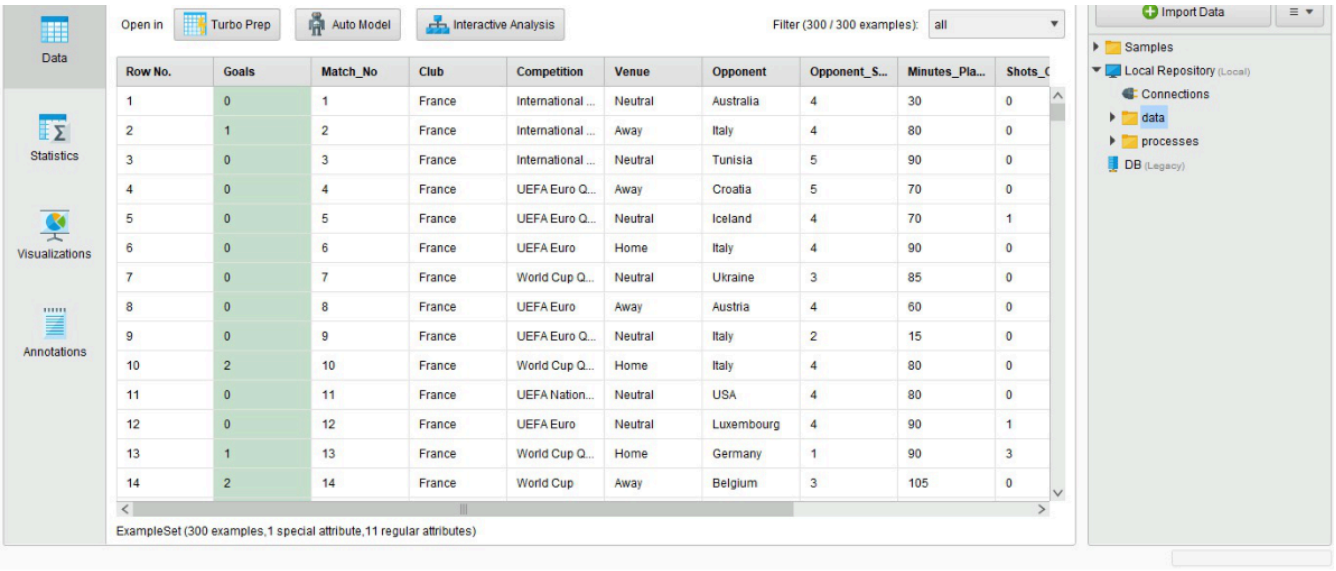
Predicting Kylian Mbappé's Goals in Altair AI Studio

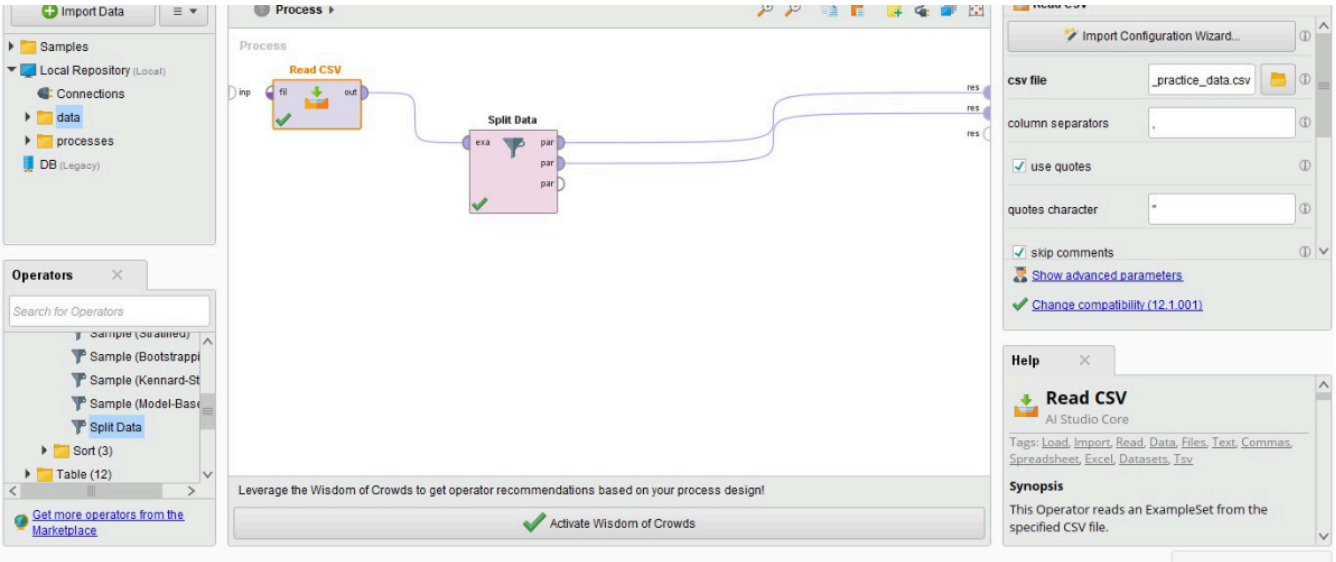
Goal: predict the Goals column from the other available match data. The workflow starts by splitting the data, then uses a Decision Tree to predict the number of goals.

Step 1: Split the Data

The dataset contains 300 international match examples. Read CSV imports the data, and Split Data divides the matches into 70% training data (210 examples) and 30% testing data (90 examples). The training data is used to build the model, while the testing data is kept separate to check how well the model predicts goals.



Imported Mbappé match data.



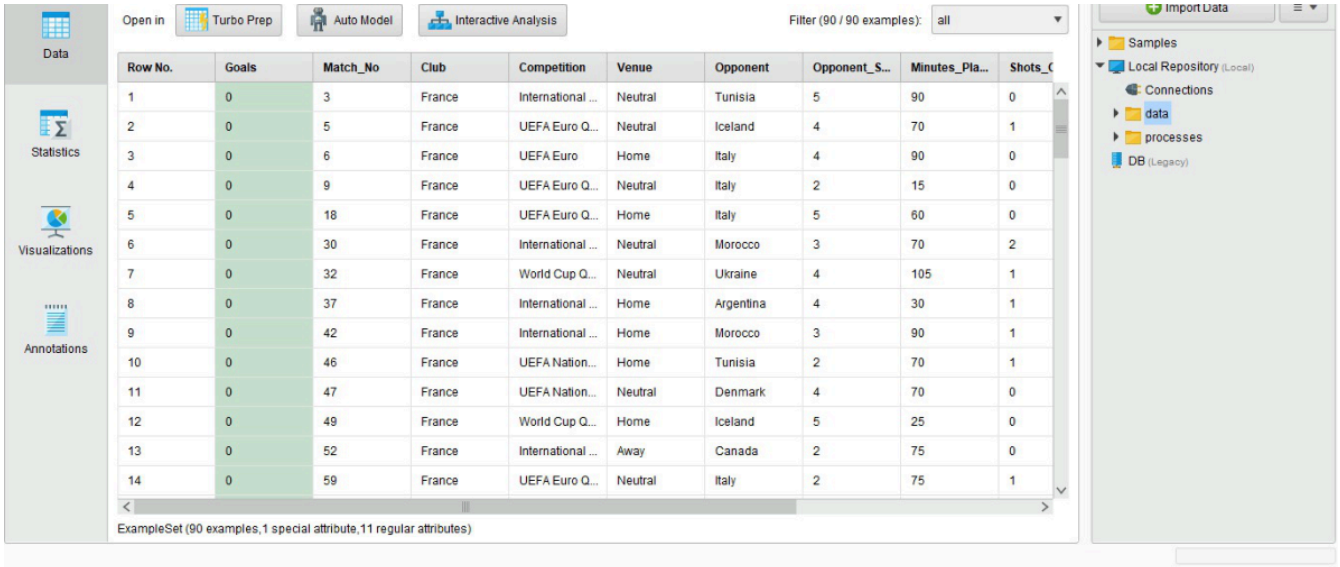
Split Data design and its parameters.

Training and Testing Data



Row No.	Goals	Match_No	Club	Competition	Venue	Opponent	Opponent_S...	Minutes_Pla...	Shots_(
1	0	1	France	International ...	Neutral	Australia	4	30	0
2	1	2	France	International ...	Away	Italy	4	80	0
3	0	4	France	UEFA Euro Q...	Away	Croatia	5	70	0
4	0	7	France	World Cup Q...	Neutral	Ukraine	3	85	0
5	0	8	France	UEFA Euro	Away	Austria	4	60	0
6	2	10	France	World Cup Q...	Home	Italy	4	80	0
7	0	11	France	UEFA Nation...	Neutral	USA	4	80	0
8	0	12	France	UEFA Euro	Neutral	Luxembourg	4	90	1
9	1	13	France	World Cup Q...	Home	Germany	1	90	3
10	2	14	France	World Cup	Away	Belgium	3	105	0
11	0	15	France	World Cup Q...	Away	Belgium	2	90	0
12	0	16	France	UEFA Nation...	Neutral	Spain	4	70	1
13	0	17	France	World Cup	Neutral	Netherlands	5	90	1
14	0	19	France	World Cup Q...	Home	Tunisia	2	90	0

Training data (210 examples, 70%).

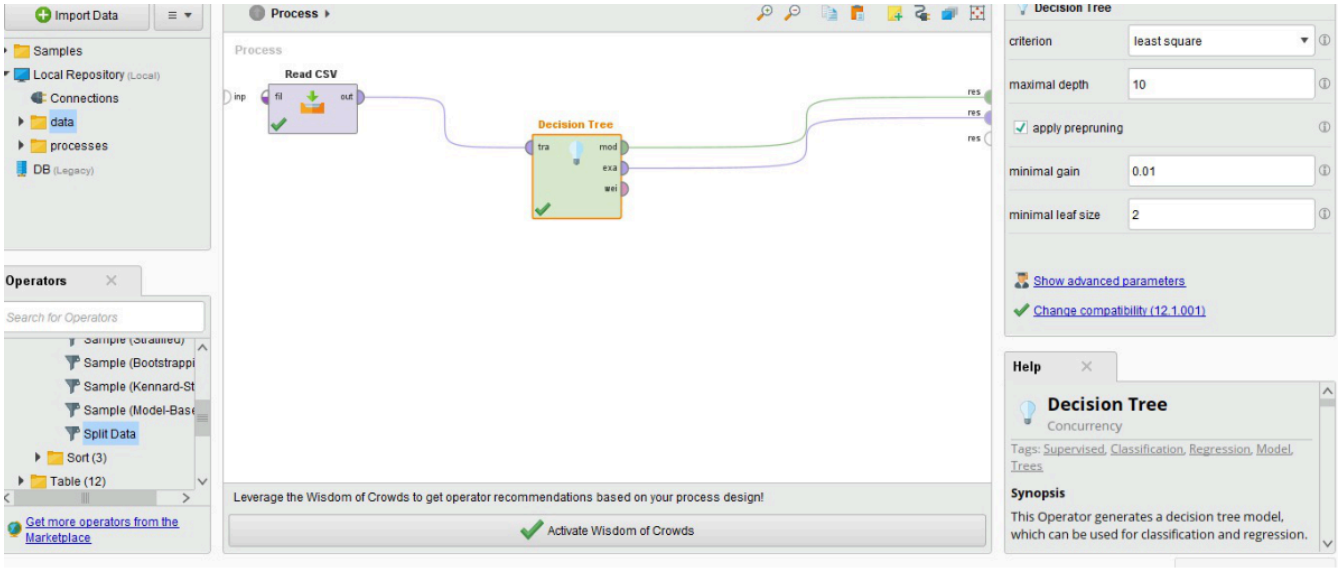


Row No.	Goals	Match_No	Club	Competition	Venue	Opponent	Opponent_S...	Minutes_Pla...	Shots_(
1	0	3	France	International ...	Neutral	Tunisia	5	90	0
2	0	5	France	UEFA Euro Q...	Neutral	Iceland	4	70	1
3	0	6	France	UEFA Euro	Home	Italy	4	90	0
4	0	9	France	UEFA Euro Q...	Neutral	Italy	2	15	0
5	0	18	France	UEFA Euro Q...	Home	Italy	5	60	0
6	0	30	France	International ...	Neutral	Morocco	3	70	2
7	0	32	France	World Cup Q...	Neutral	Ukraine	4	105	1
8	0	37	France	International ...	Home	Argentina	4	30	1
9	0	42	France	International ...	Home	Morocco	3	90	1
10	0	46	France	UEFA Nation...	Home	Tunisia	2	70	1
11	0	47	France	UEFA Nation...	Neutral	Denmark	4	70	0
12	0	49	France	World Cup Q...	Home	Iceland	5	25	0
13	0	52	France	International ...	Away	Canada	2	75	0
14	0	59	France	UEFA Euro Q...	Neutral	Italy	2	75	1

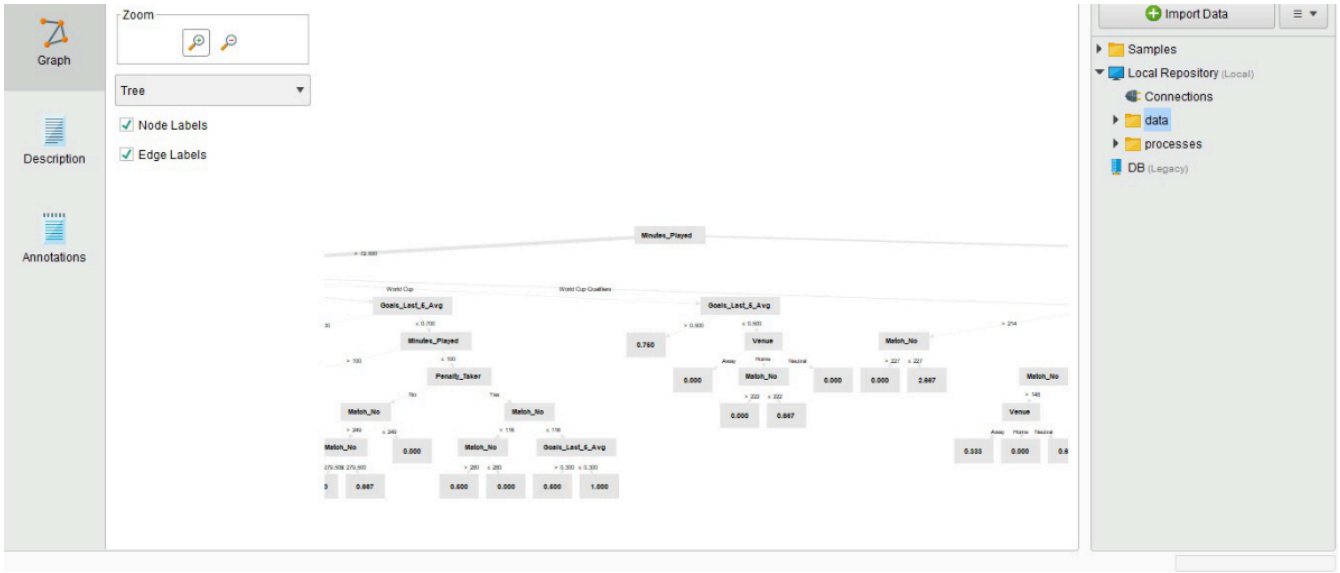
Testing data (90 examples, 30%).

Step 2: Decision Tree

The Decision Tree is trained using the training part of the split data. It learns patterns from the available match columns and uses them to predict the Goals value. The dataset includes details such as opponent, venue, minutes played and shots on target.



Process design with the match data connected to the Decision Tree operator.



Resulting decision tree (Graph view). Each box shows a split in the data, and the end boxes show predicted Goals values.

The 300 match examples were divided into 70% training data (210 matches) and 30% testing data (90 matches). The training data was used to build a Decision Tree model to predict Mbappé?s Goals value, and the testing data was kept separate for evaluation.